

Topic 9: Sound and Music Description

Audio features, sound collections and Freesound classification

Digital Signal Processing for Sound and Music · Topic 9

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Lecture roadmap

Spectral features support decisions about sounds and sound collections

1 Frame features

Energy, centroid, MFCC,
pitch salience and chroma

2 Time and statistics

Onsets, pitch contours and
recording summaries

3 Sound collections

Similarity, clustering and
classification

4 Freesound examples

SVM classification and PCA
exploration

5 Music description

Musical facets and the role
of context

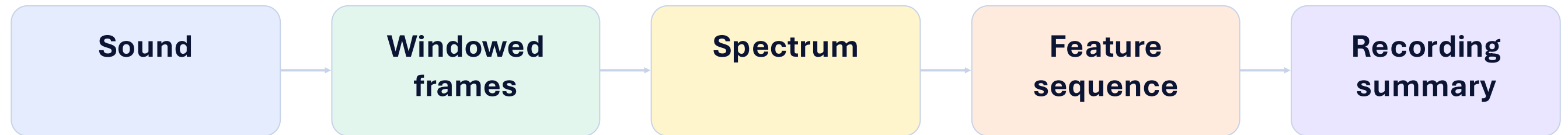
6 Exercise 9

Coherent data and
explained experiments

Audio features

A feature reduces selected properties of a sound to numerical values

- Each frame gives a local description of the sound.
- A feature sequence describes how that property changes over time.
- Statistics can summarise a sequence for one recording.



The representation should retain the properties needed for the intended task.

Levels of sound and music description

Higher levels require interpretation and context beyond a measured spectrum

Level	Examples	What connects the levels
Physical	Frequency, spectrum, duration, intensity	Measurements from the signal
Sensorial	Pitch, timbre, loudness	Perceptual models of the measurements
Perceptual	Intervals, beat, spectral envelope	Relationships over time or between components
Formal	Melody, key, meter, instrumentation	Models of musical organisation
Cognitive	Style, emotion, semantic concepts	Listeners, culture and listening context

Essentia descriptor families

Different algorithms capture different aspects of a recording

Family	Examples	Typical question
Spectral	Centroid, MFCC, HFC, contrast	How does the spectral shape change?
Time and dynamics	RMS, loudness, attack time, ZCR	How strong or abrupt is the sound?
Tonal	Pitch, pitch salience, HPCP	Which pitch structures are present?
Rhythm and events	Flux, onsets, beat, tempo	When do events occur?
Musical models	Key, chords, structural boundaries	How does the music organise its material?

Energy, RMS and loudness

Energy and RMS measure signal level, while perceived loudness also depends on the sound

A frame of M samples

$$E = \sum_{n=0}^{M-1} x[n]^2$$

$$RMS = \sqrt{\frac{E}{M}}$$

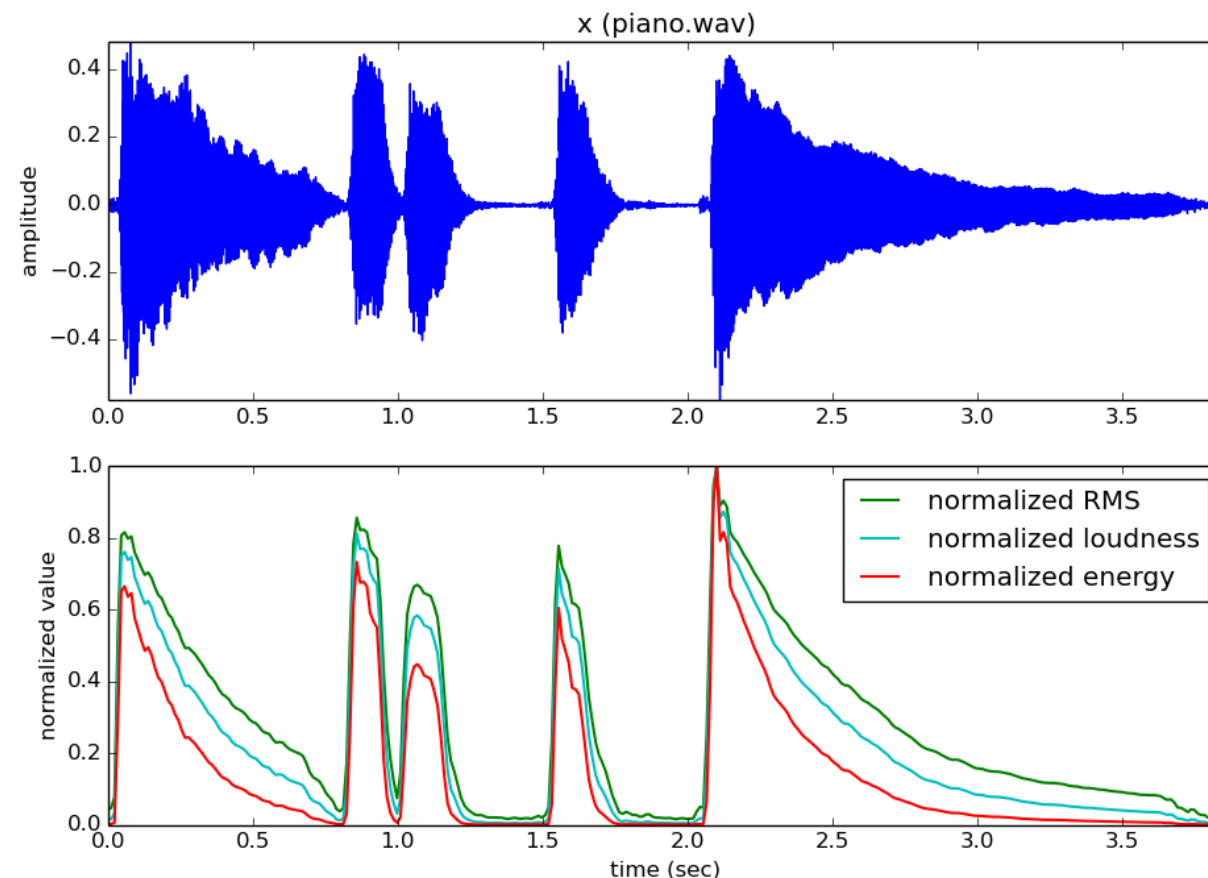
$$L_{proxy} = E^{0.67}$$

Interpretation

- Doubling amplitude multiplies energy by 4 and RMS by 2.
- The power law is a simple loudness proxy used in this lecture.
- Absolute perceived loudness requires calibration and frequency-dependent modelling.

Example: level measures in a piano recording

The measures follow the same attacks but use different numerical mappings



Original 9T1 example: piano.wav

How to read it

- Top: the piano waveform.
- Bottom: normalised energy, RMS and loudness proxy.
- Normalisation lets us compare shapes, not absolute units.
- A softer tail can remain visible after compression.

Spectral centroid

A weighted mean locates the centre of a magnitude spectrum

Frequency-weighted average

$$C = \frac{\sum_{k=0}^{K-1} f[k] A[k]}{\sum_{k=0}^{K-1} A[k]}$$

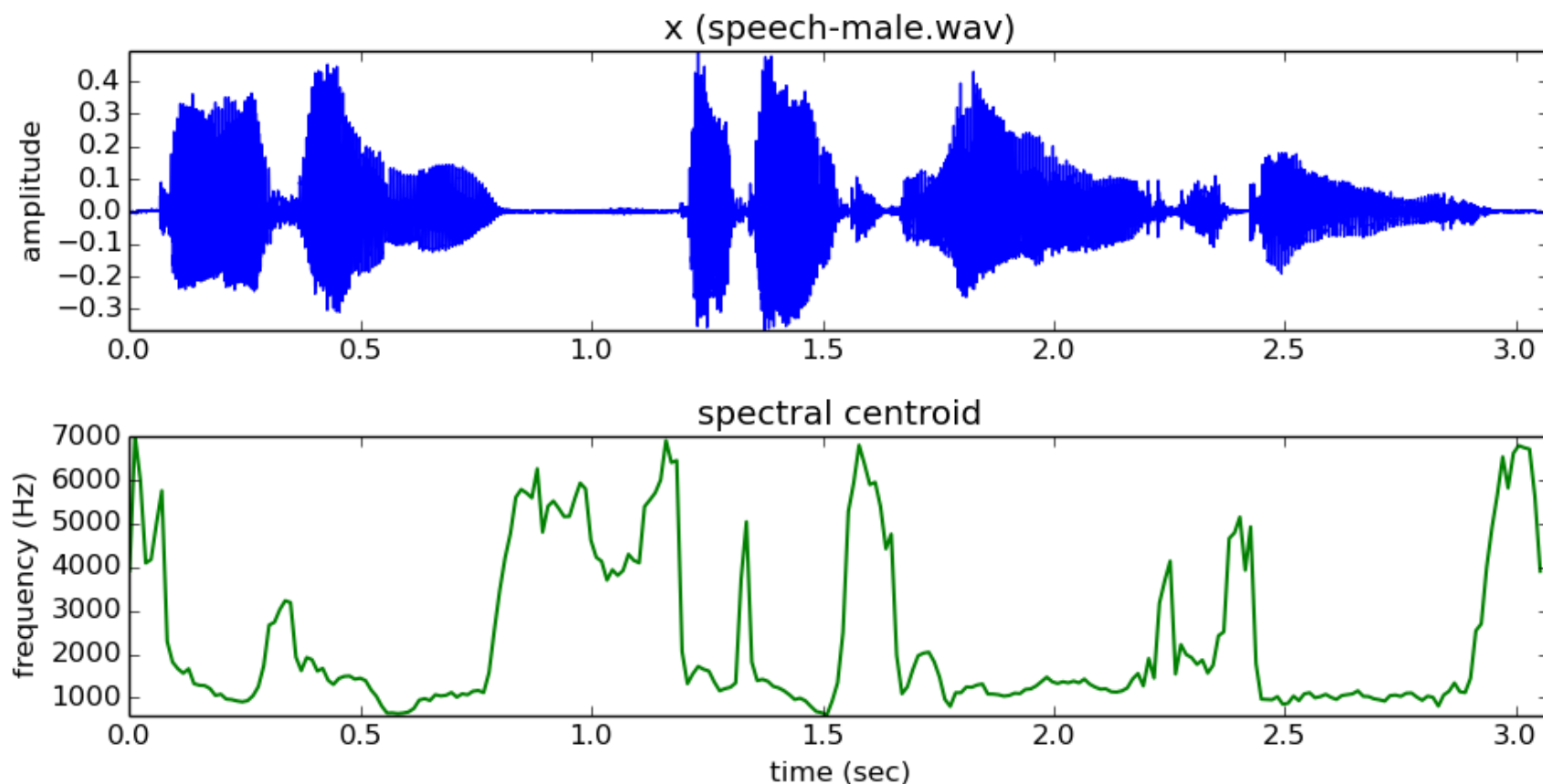
$A[k]$: linear magnitude; K : number of bins.
 $f[k]$: bin frequency in Hz.

A small example

- 500 Hz has weight 3.
- 2000 Hz has weight 1.
- Centroid = $(500 \times 3 + 2000 \times 1) / 4 = 875$ Hz.
- Higher values often correlate with greater brightness.

Example: spectral centroid in speech

Centroid follows spectral balance even when waveform amplitude remains similar



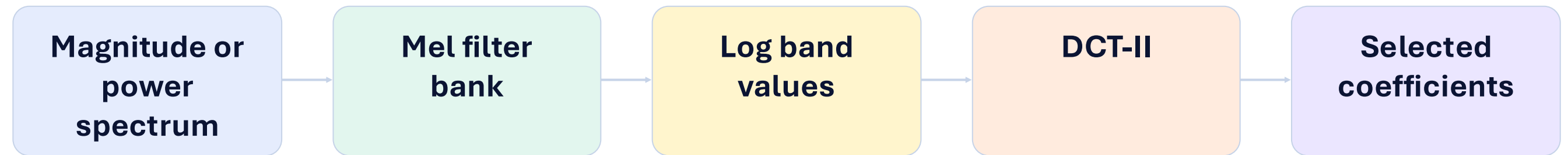
Original 9T1 example: speech-male.wav

What changes?

- Centroid measures spectral centre in Hz.
- Some consonants emphasise high frequencies.
- Speech sounds change the curve over time.
- This is spectral balance, not voice pitch.

Mel-frequency cepstral coefficients

MFCCs describe the shape of a compressed spectrum on a perceptual frequency scale

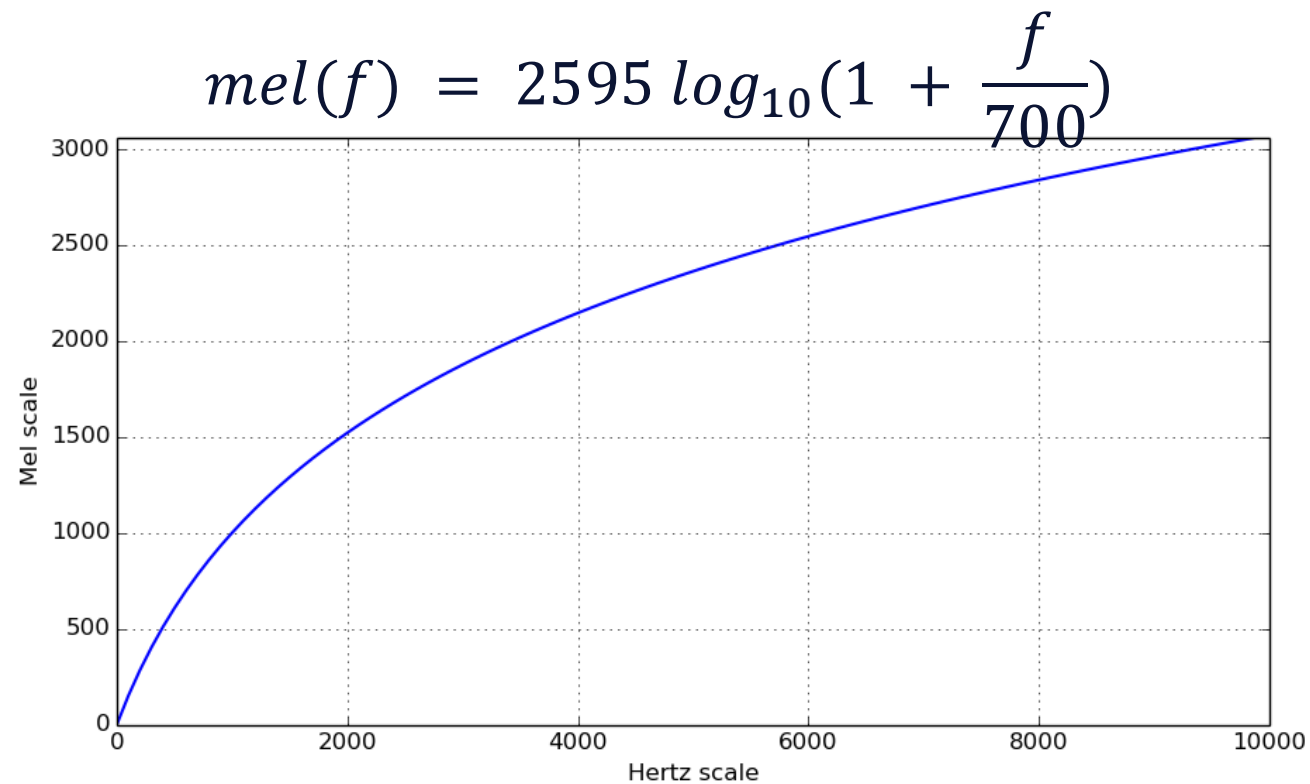


Meaning of the stages

- Mel bands combine neighbouring spectral bins into broader frequency regions.
- Log compression reduces the effect of large amplitude differences.
- The DCT represents the log band pattern with cosine-shaped weights.

The mel frequency scale

Equal mel steps correspond to smaller frequency intervals at low frequencies



Original mel-scale illustration from 9T1

Intuition

- Low-frequency filters are relatively narrow in Hz.
- Higher-frequency filters span wider Hz ranges.
- This is an auditory-inspired scale, not a complete hearing model.

What one MFCC coefficient represents

Each coefficient measures agreement with a cosine pattern across the log mel bands

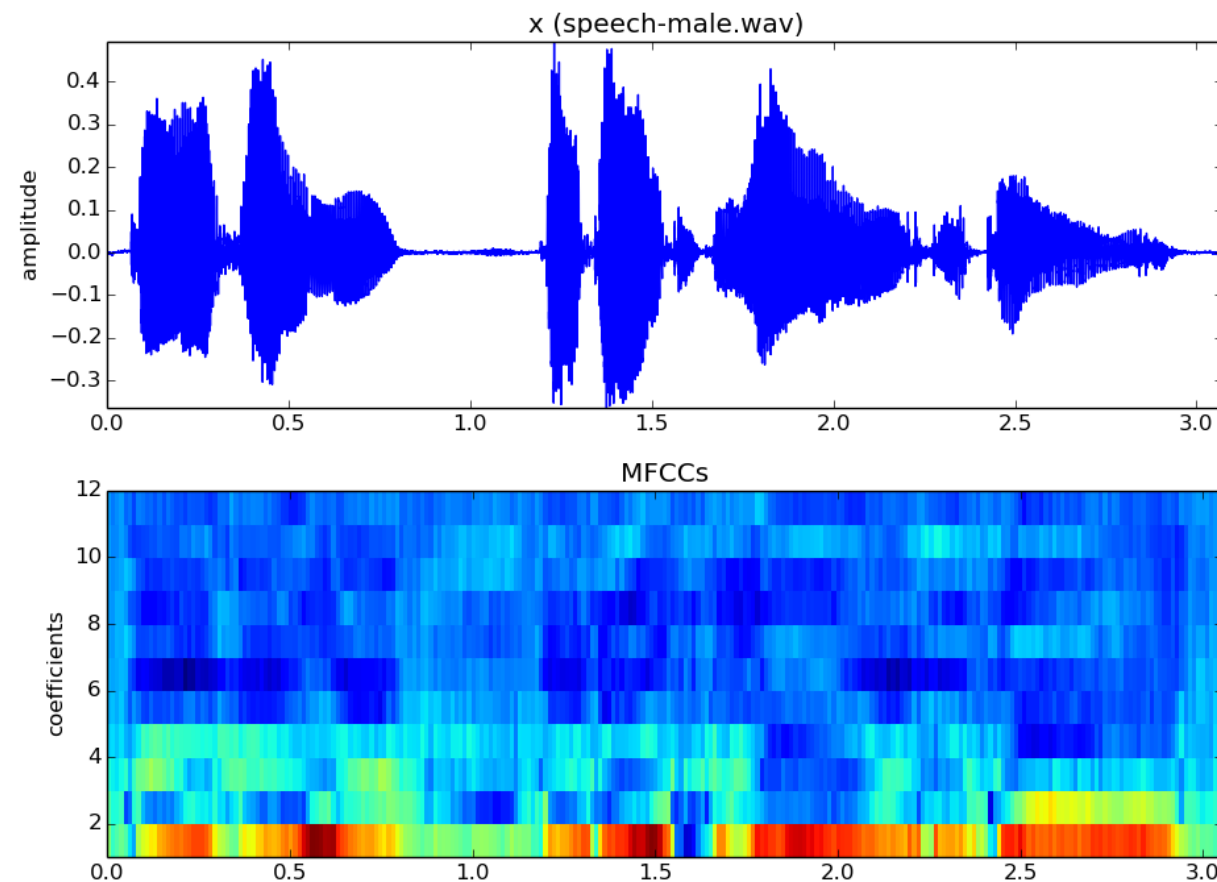
$$c_r = a_r \int_{b=0}^{B-1} L_b \cos\left(\frac{\pi r (b + \frac{1}{2})}{B}\right)$$

B: number of mel bands L[b]: log band value
r: coefficient index a[r]: DCT normalisation

- c[0] summarises the average log band level.
- Low-order coefficients describe broad spectral patterns.
- Positive and negative weights compare regions. A negative coefficient is a signed comparison, not negative sound energy.

Example: MFCCs in speech

The heatmap shows a vector of coefficients for every frame



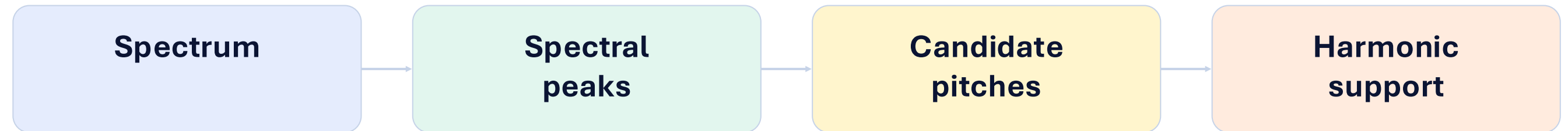
Original 9T1 MFCC analysis of speech

Reading the map

- Horizontal position: time.
- Vertical position: coefficient index.
- Colour: coefficient value, which can be signed.
- Changes in the map describe changes in spectral shape.

Pitch salience functions

A candidate pitch gains support when peaks appear near its harmonics



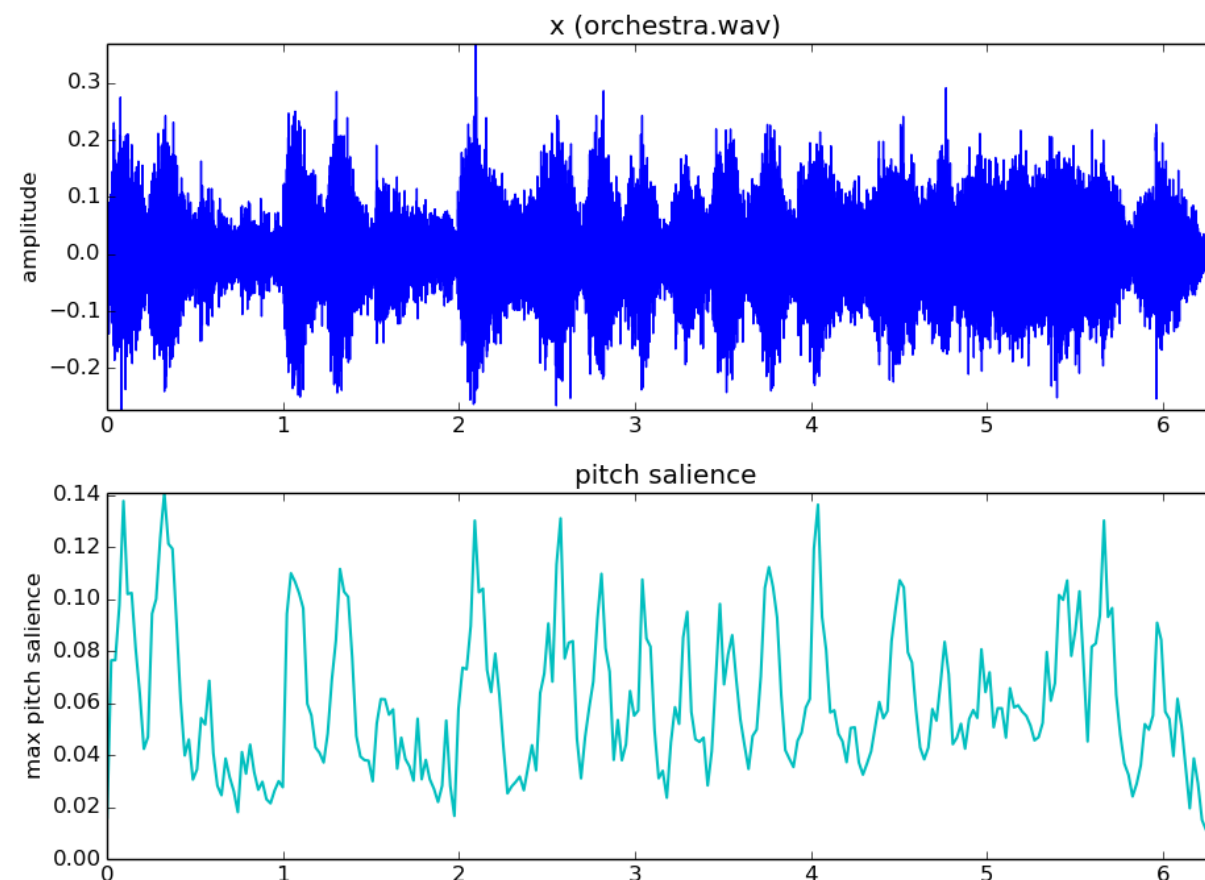
$$S(f_0) \approx \int_{h=1}^H w_h A(h f_0)$$

Illustrative harmonic-support score.
 $A(h f_0)$ denotes peak evidence near harmonic h .

- A weak or missing fundamental can still receive harmonic support.
- Several candidate pitches can be salient in one polyphonic frame.

Example: maximum pitch salience in an orchestra

A single value summarises the strongest pitch evidence in each frame



Original 9T1 maximum pitch-salience plot: orchestra.wav

Peak strength

- The curve tracks the strongest pitch support.
- Its height indicates strength, not f_0 .
- The peak location gives a candidate f_0 .
- Several pitches may still be plausible.

Chroma and harmonic pitch class profiles

Pitch classes group notes across octaves and describe tonal content

Weighted peak contributions

$$HPCP[c] = \int_{p=1}^P w(c, f_p) A_p^2$$

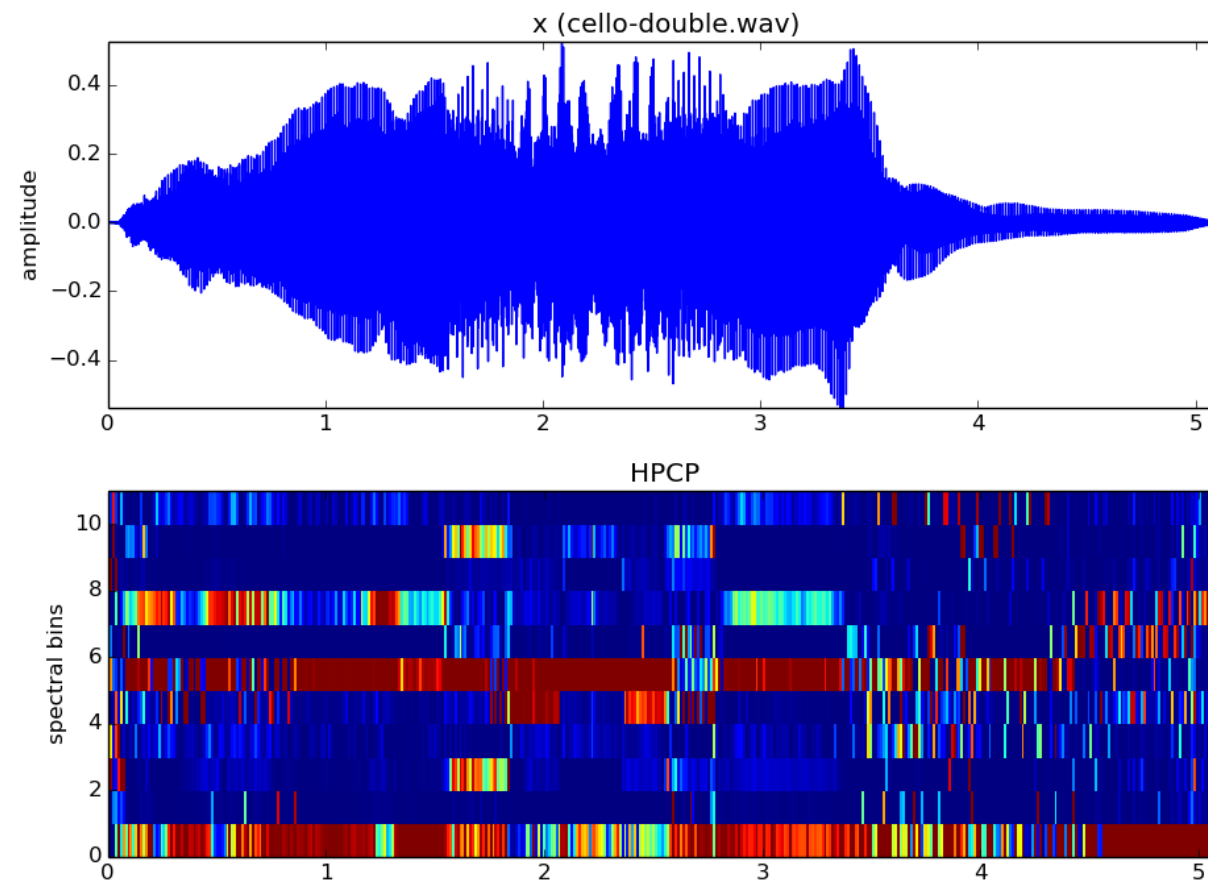
Each peak contributes to nearby pitch-class bins.
Tuning and weighting define those contributions.

Octave folding

- A at 110, 220 and 440 Hz contributes to the same pitch class.
- An HPCP can use 12 bins or a finer multiple of 12.
- Octave folding helps compare harmony but loses pitch register.

Example: HPCP analysis of a cello recording

Active pitch-class bins reveal tonal changes across the recording



Original 9T1 example: cello-double.wav

Reading the map

- Columns summarise pitch-class energy.
- Octave versions reinforce the same bins.
- Notes and harmonics can activate several bins.
- Key and chord estimates need another model.

Spectral change and onset detection

An onset detector looks for a rise or change, then selects event times

One positive-flux definition

$$F[\ell] = \int_{k=0}^{K-1} \max(0, A[\ell, k] - A[\ell - 1, k])$$

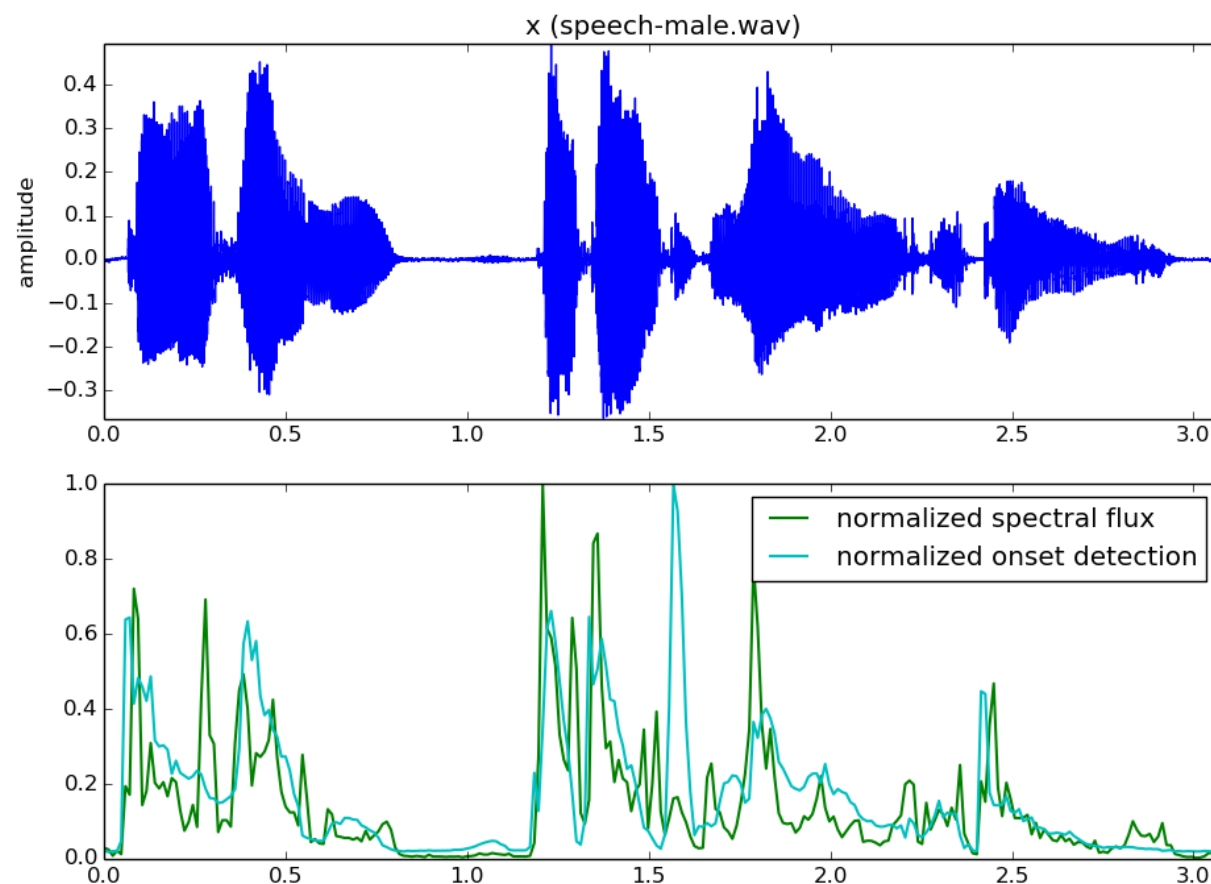
Only magnitude increases contribute here.
Other flux algorithms use different norms or normalisations.

Event decisions

- Flux compares consecutive spectra.
- HFC weights higher frequencies more strongly. Its changes can emphasise attacks.
- Peak selection and thresholds turn a detection curve into onset times.

Example: spectral change in speech

Detection-function peaks mark candidate changes that need thresholding and interpretation



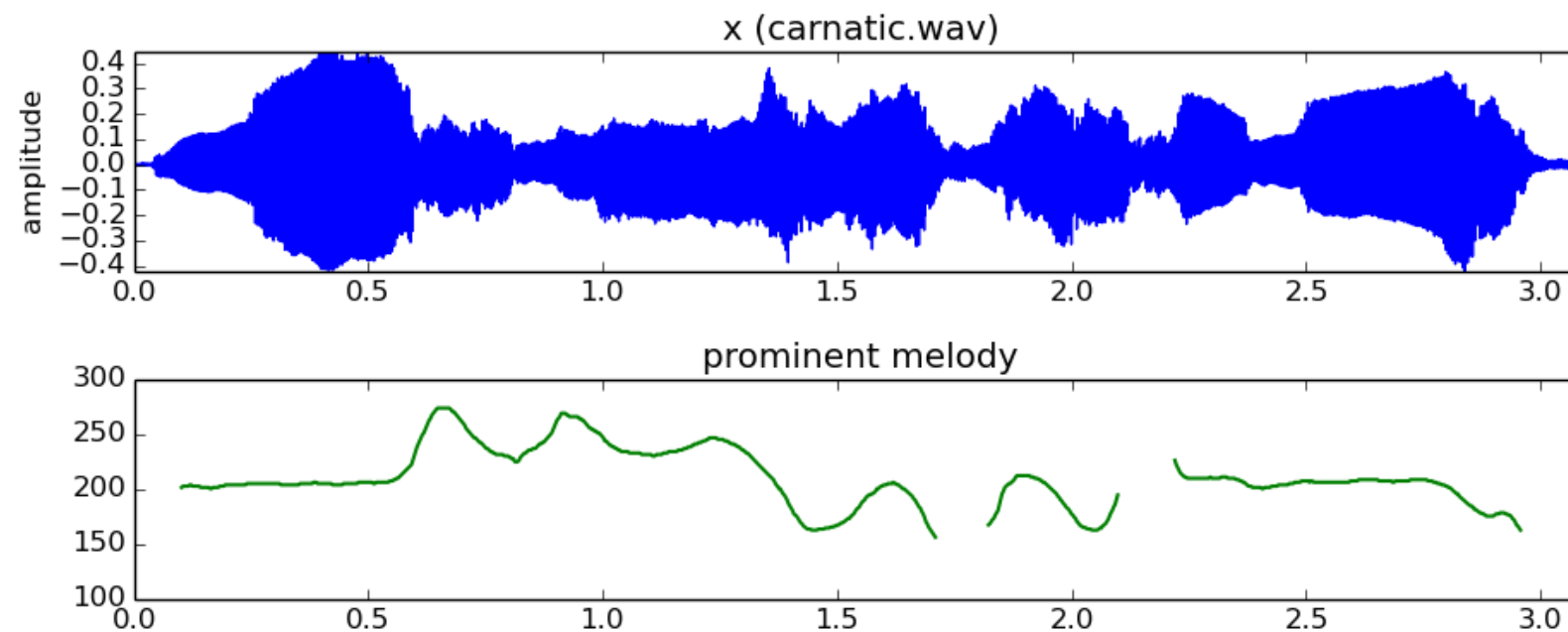
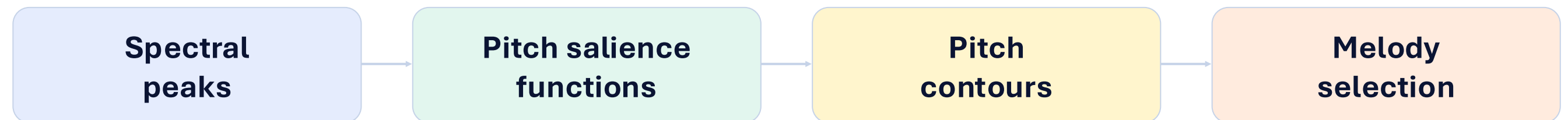
Original 9T1 spectral-flux and onset example

What the curves show

- The waveform gives the temporal reference.
- Lower curves: normalised detection functions.
- Peaks can mark attacks or spectral changes.
- One event may generate several nearby peaks.

Predominant pitch across several frames

Temporal continuity helps select a melody contour among competing candidates



Interpretation

- Several frames support a continuous contour.
- Unvoiced regions need an explicit convention.
- The chosen contour can differ from the bass or loudest peak.

Statistics of a feature sequence

A recording summary aggregates T frame values of one descriptor

$$\mu = \frac{1}{T} \int_{t=0}^{T-1} v[t]$$

$$\sigma^2 = \frac{1}{T} \int_{t=0}^{T-1} (v[t] - \mu)^2$$

$$\gamma_1 = \frac{1}{T} \int_{t=0}^{T-1} \left(\frac{v[t] - \mu}{\sigma} \right)^3$$

What survives?

- Mean: typical feature value.
- Variance: amount of variation.
- Skewness: asymmetry around the mean.
- These statistics discard the order of the frames.

Frame sequences and recording summaries

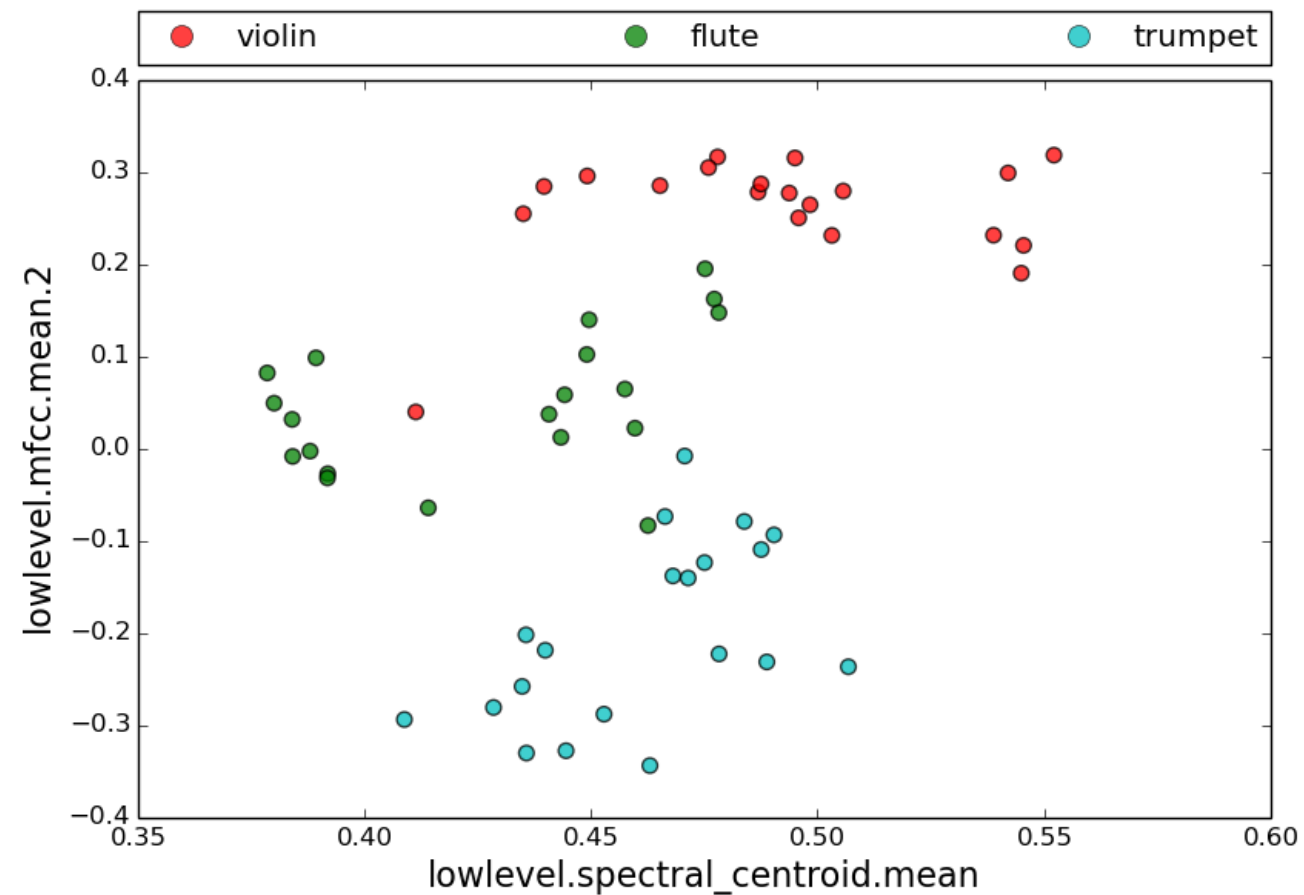
Identical statistics can describe different temporal behaviours

Representation	Useful information	Limits
Frame sequence	Evolution, attacks, sustained sections	Requires alignment or a temporal model
Mean per descriptor	Average spectral or tonal properties	Variation and frame order
Mean and variance	Average plus amount of variation	Order and event structure
Segments and trajectories	Local patterns and temporal organisation	Depends on segmentation choices

Example: a crescendo and a decrescendo can have the same mean and variance.

A sound collection in descriptor space

One recording becomes one vector after feature extraction and aggregation



Original 9T2 violin, flute and trumpet example

Each point is a sound

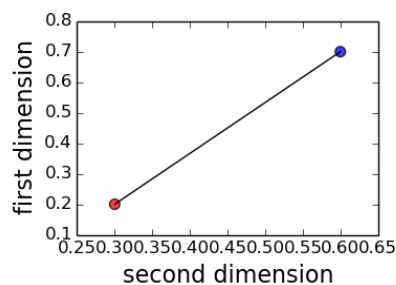
- Two descriptors define the axes.
- Colours show known instrument labels.
- Compact groups share measured properties.
- Overlap can reflect missing information.

Euclidean distance between descriptor vectors

Distance measures proximity in the chosen numerical representation

$$d(p, q) = \sqrt{\sum_{i=1}^D (q_i - p_i)^2}$$

p and q contain the same D descriptors in the same order.



Meaning of similarity

- A small distance means similar selected feature values.
- Feature scaling changes which differences dominate.
- Timbral similarity, instrument identity and musical similarity can disagree.

Feature scaling before measuring distance

Each dimension should contribute on an intentional numerical scale

Workflow	Scaling in the supplied code	Consequence
E9 k-means	SciPy whiten: x divided by its standard deviation	Descriptors have unit variance without mean centring
E9 k-NN	Raw selected descriptor values	Large numerical ranges can dominate distance
Freesound SVM example	z-score using training mean and standard deviation	Test and query use the training transformation

$$z_i = \frac{x_i - \mu_{train,i}}{\sigma_{train,i}}$$

Similarity, clustering and classification

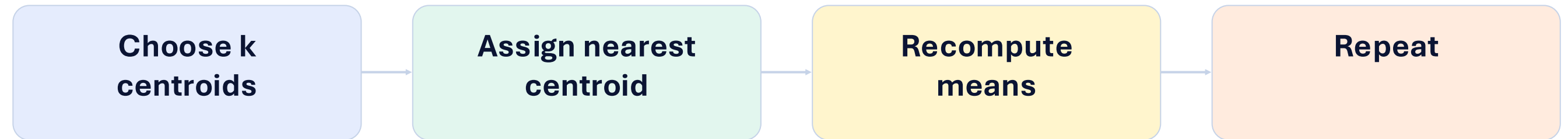
These tasks use a feature space to answer different questions

Task	Inputs	Output
Similarity retrieval	A query and reference sounds	A ranked list of nearby sounds
Clustering	Unlabelled vectors and a grouping criterion	Groups defined by the data geometry
Classification	Labelled training examples and a query	One of the available class labels

Known labels can evaluate a clustering result without taking part in the clustering.

K-means clustering

Assignments and centroids update until the grouping stabilises

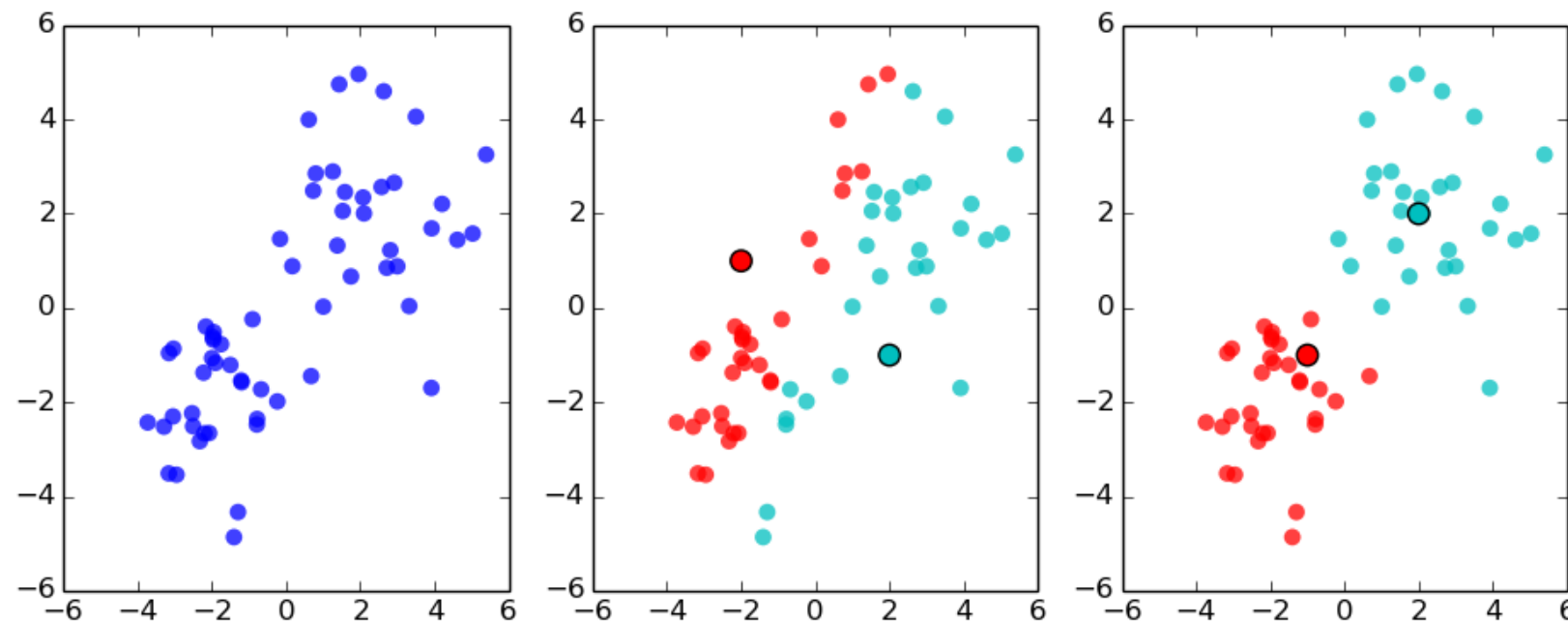


$$\min \int_{j=1}^S \|z_j - \mu_{c(j)}\|^2$$

- The objective measures squared distance within clusters.
- A centroid is a mean vector, not necessarily a real sound.
- Different initialisations can produce different results.

Example: grouping a sound collection

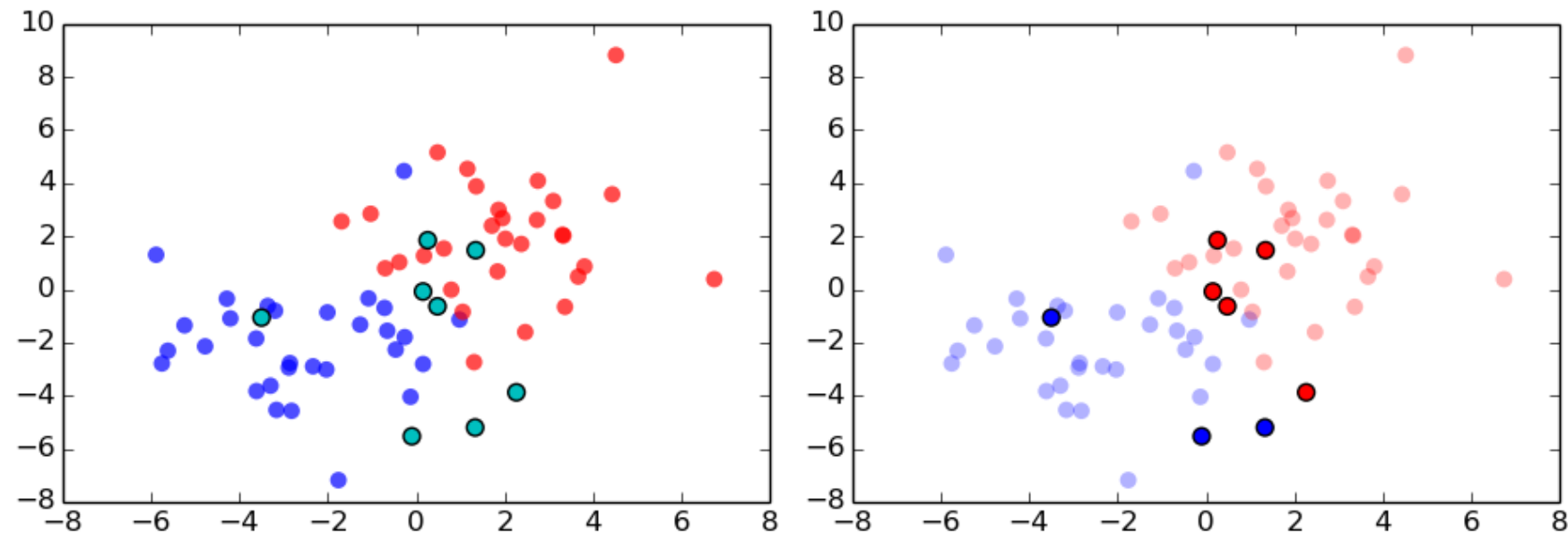
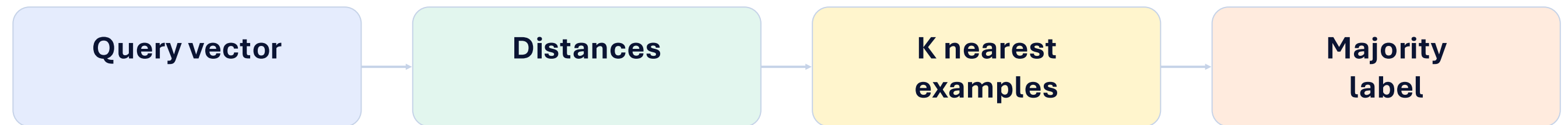
Cluster membership follows feature geometry rather than instrument names



Left: ungrouped points. Middle: initial assignments. Right: updated groups and centroids. Geometric groups need not match instrument labels.

K-nearest-neighbour classification

A query receives the most frequent label among its K nearest reference sounds



Role of K

- $K = 1$ follows the closest example.
- Larger K averages over a wider neighbourhood.
- The prediction depends on the descriptors and available labels.

A worked k-NN voting example

The nearest individual sound and the majority label can disagree

Neighbour rank	Distance	Known class
1	0.20	Trumpet
2	0.25	Violin
3	0.28	Violin

K = 1 gives Trumpet. K = 3 gives Violin by two votes to one.

Evaluation of classification

Performance should describe predictions on sounds excluded from training

Data separation

- Training sounds define the model and scaling.
- Held-out sounds estimate generalisation.
- Near duplicates from one pack can make a random split too easy.

Reading errors

- A confusion matrix shows which classes mix.
- Accuracy summarises correct predictions but can hide uneven performance.
- Listening to errors helps identify missing features or faulty labels.

Freesound classification notebooks

The exercise and repository examples share descriptors but use different methods

Material	Main methods	Role
E9 exercise	2D plots, k-means and k-NN	Explain data choice and descriptor-space decisions
freesound-classification-core.ipynb	SVM with an RBF kernel	Prepare, standardise, train, evaluate and predict
freesound-classification-visualization.ipynb	PCA and interactive playback	Inspect the trained result and listen to points

[Open the Freesound classification core](#)

[Open the Freesound classification visualization](#)

Freesound SVM example: configuration

Three training labels define the choices available to the classifier

Setting	Value in the notebook	Interpretation
Training collections	Violin pizzicato, naobo, trumpet single notes	20 sounds per class, durations 0–3 seconds
Descriptor selection	Indices 0–12	13 scalar dimensions, including vector components
Data split	80% train, 20% test, stratified, seed 42	Saved run has 48 train and 12 test sounds
Classifier	RBF SVM, C = 10, gamma = scale	Nonlinear classification in standardised space
Query	Viola, pizzicato, duration 0–3 seconds	Viola is absent from the training labels

Support vector machines

An SVM learns class boundaries from labelled training examples

Intuitive view

- A linear SVM seeks a separating boundary with a wide margin.
- Support vectors constrain the fitted boundary.
- An RBF kernel allows curved boundaries in the original feature space.

Controls in the example

- C controls the trade-off between training errors and the margin.
- Gamma sets the locality of the RBF response.
- Feature scaling changes the geometry, so the example standardises first.

Freesound SVM example: saved evaluation

The saved run correctly predicts all 12 held-out examples

True class / predicted class	Naobo	Trumpet	Violin
Naobo	4	0	0
Trumpet	0	4	0
Violin	0	0	4

Saved train accuracy: 100% Saved test accuracy: 100%
A test set of 12 sounds gives limited evidence about new recording conditions.

Freesound query: a viola receives the violin label

The classifier must choose among naobo, trumpet and violin

Saved notebook result

Query: viola pizzicato

Freesound ID: 374382

Prediction: violin

Reported model probability: 93.9%

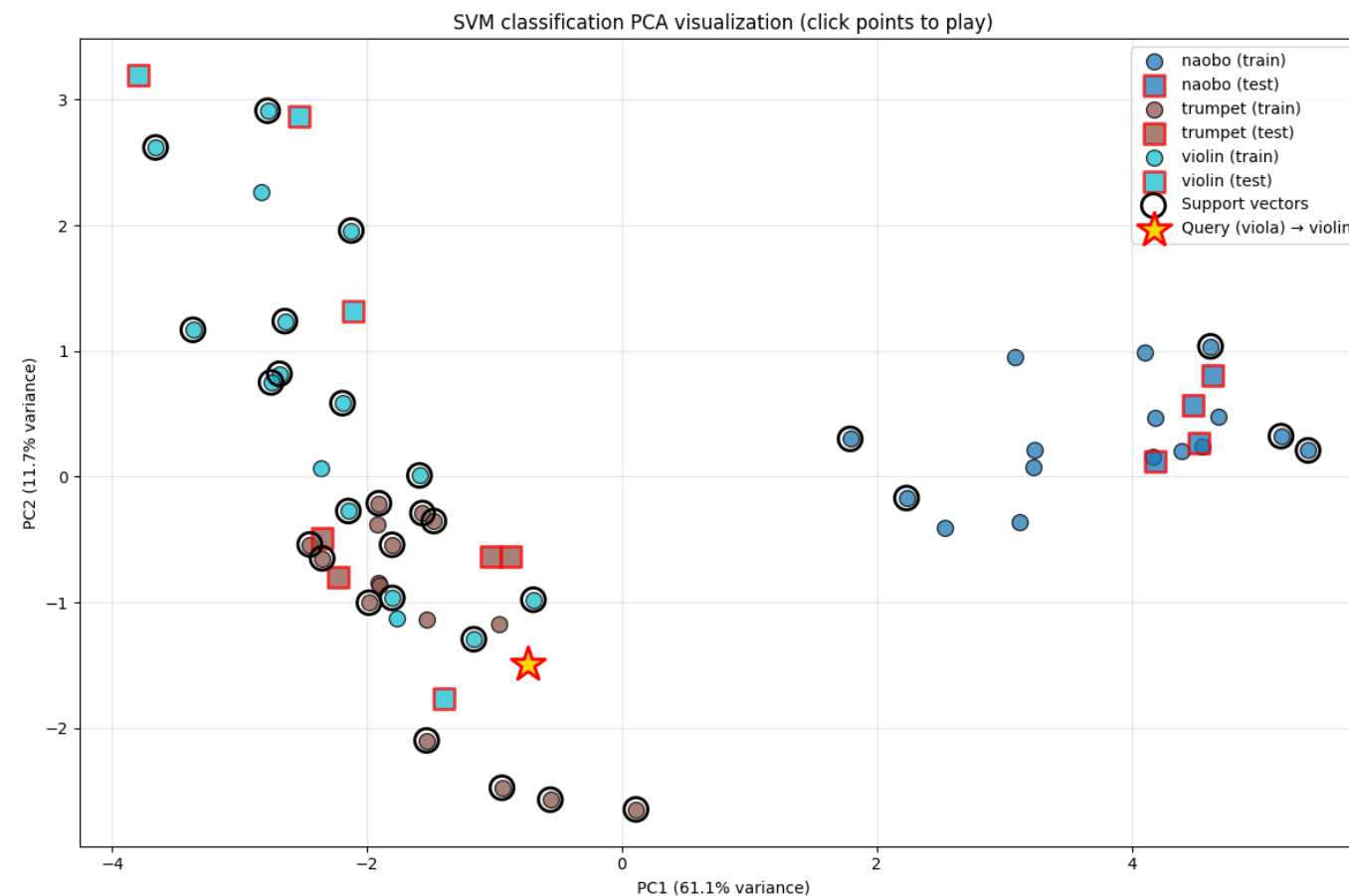
[Listen to the query on Freesound](#)

What it means

- The training set has no viola class.
- Shared timbral properties can support a violin decision.
- The probability is conditional on the available classes and model assumptions.

Freesound example: PCA exploration and listening

The 2D view is a projection of the descriptor space used by the SVM



Saved PCA output from the visualization notebook

Markers and axes

- Circles: training. Squares: test.
- Rings: support vectors. Star: viola query.
- These axes show 72.8% of variance.
- The SVM uses all 13 dimensions.

Description of music recordings

Musical concepts need models that connect audio evidence with musical organisation

Musical facet	Examples	Evidence and interpretation
Timbre and instrumentation	Instrument roles, ensemble, production	Spectral and temporal cues plus learned/contextual models
Melody and harmony	Motifs, tonic, chords, key, raga, makam	Pitch relationships, repetition and musical context
Rhythm	Beat, downbeat, meter, rhythmic patterns	Event timing plus periodicity and grouping
Structure	Phrases, sections, movements	Repetition, contrast and longer-term organisation

Description of music collections

Similarity depends on the musical facet and the purpose of the comparison

Musical similarity

- Recordings can share rhythm while differing in instrumentation.
- Melodic similarity may survive changes of key or tempo.
- Structural similarity requires relationships across a recording.

Classification and context

- Genre and style labels depend on musical communities.
- One descriptor space may emphasise production over composition.
- Listening and metadata help interpret what a model has captured.

Exercise 9: sound and music description

Four parts connect coherent collections with explained machine-learning decisions

1 • Collections

Download three instrument collections and their descriptors.

2 • Descriptor pair

Compare 2D plots and justify a useful pair.

3 • K-means

Compare descriptor subsets and clustering stability.

4 • K-NN

Classify queries and explain their nearest neighbours.

Lab 9, Part 1: coherent Freesound collections

Search and listening decisions determine what the feature space will represent

Collection requirements

- Choose 3 of: violin, guitar, bassoon, trumpet, clarinet, cello, naobo.
- Download 20 valid single-note or single-stroke sounds per instrument, under 10 seconds.
- Record sound IDs, links and the final query, tag and duration filters.

Inspection

- Compare an initial and a refined query.
- Reject wrong instruments, polyphonic passages or unsuitable effects.
- Listen to descriptor outliers before deciding whether to remove them.

Lab 9: descriptor index mapping

A vector-valued descriptor contributes several scalar dimensions

Indices	Descriptor components	What they help describe
0	Mean spectral centroid	Spectral balance or brightness
1	Mean dissonance	Sensory roughness model
2	Mean HFC	Emphasis on high-frequency content
3	Mean log attack time	Speed of the amplitude attack
4	Mean inharmonicity	Deviation from harmonic frequency relationships
5–10	Mean spectral contrast components 0–5	Peak–valley contrast in spectral bands
11–16	Mean MFCC components 0–5	Log mel spectral shape

Lab 9, Parts 2 and 3: descriptor experiments

A useful feature set separates classes while remaining stable across runs

Stage	Required comparisons	What to explain
2D descriptor plots	At least 4 descriptor pairs	Within-class compactness, between-class separation and outliers
K-means	At least 3 descriptor subsets, nCluster = 3	Which confusions persist and why
Repeated clustering	At least 5 runs of the best subset	Mean and range of the reported clustering score
Error inspection	Listen to one wrongly grouped sound	Use at least two descriptors to justify ambiguity

E9 labels clusters by majority vote after fitting. Its score summarises cluster purity.

Lab 9, Part 4: classification and interpretation

At least five experiments should connect predictions to descriptor-space evidence

Experimental record

- Report query ID/link, descriptor subset, K and predicted class.
- Compare a small K with a larger K.
- Include at least one query outside the three training classes.

Explanation

- Listen to the nearest neighbours in one expected and one surprising case.
- Compare distances to the nearest examples from competing classes.
- Explain how articulation, pitch register or recording conditions could change the result.

What to remember from Topic 9

Features describe selected evidence that a task can use

1 Features

Choose descriptors that capture the sound properties you need.

2 Time

Frame sequences retain evolution. Global moments lose order.

3 Geometry

Scaling and descriptor choice define numerical similarity.

4 Learning

Clusters group data.
Classifiers choose among known labels.

5 Examples

A clear 2D plot and a high score have limited scope.

6 Interpretation

Inspect sounds and explain predictions in musical context.

References and credits

Lecture sources, implementations and the Freesound example notebooks

- Xavier Serra: 9T1 spectral-based audio features and 9T2 sound/music description.
- E9-Sound-and-music-description.ipynb: collection, clustering and k-NN exercise.
- [Essentia algorithm reference: definitions, inputs and parameter conventions.](#)
- [Freesound classification core: SVM training, evaluation and query prediction.](#)
- [Freesound classification visualization: PCA, markers and interactive listening.](#)
- [Freesound API documentation and SciPy / scikit-learn documentation.](#)
- Original lecture slides: CC BY-NC-SA. Repository software: AGPL. Individual Freesound licences apply.